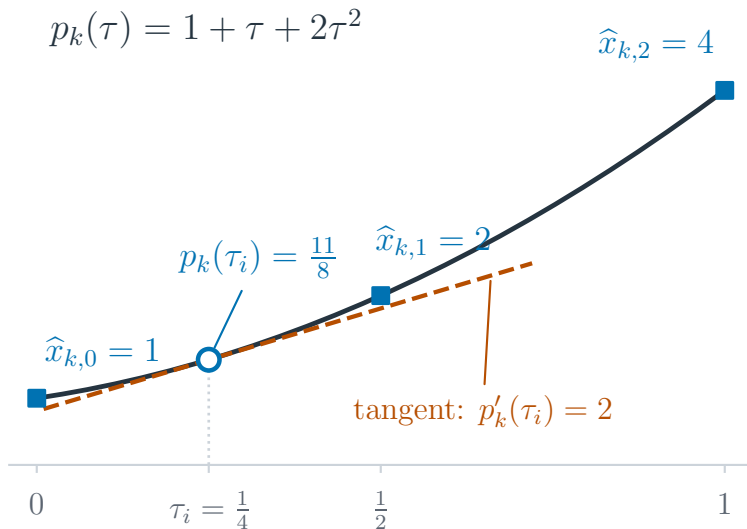


Same support values: a value from E , a slope from D



Squares: stored states at $\sigma = (0, \frac{1}{2}, 1)$.

Open circle: evaluation at $\tau_i = \frac{1}{4}$.

Evaluate with E

$$\sum_r E_{ir} \hat{x}_{k,r} = p_k(\tau_i) = \frac{11}{8}$$

The state supplied to the ODE

Differentiate with D

$$\sum_r D_{ir} \hat{x}_{k,r} = p'_k(\tau_i) = 2$$



divide by $h_k = 2$

$$\frac{1}{h_k} p'_k(\tau_i) = 1$$

The slope per unit of physical time

Use $\dot{x} = u$, $t_k = 2$, $h_k = 2$, and $u_{k,i} = 1$.

$$\underbrace{\frac{1}{h_k} \sum_r D_{ir} \hat{x}_{k,r}}_{\text{polynomial slope} = 1} = \underbrace{f\left(\sum_r E_{ir} \hat{x}_{k,r}, u_{k,i}, t_k + h_k \tau_i\right)}_{\text{ODE slope} = 1}$$

The collocation constraint equates these two physical-time rates.